

Unit 1: Fundamentals of Chemistry (CO1)

Define atomic number and mass number. Give one example.

Answer: Atomic number is the number of protons in an atom's nucleus, while mass number is the sum of protons and neutrons. For example, in Carbon-12, the atomic number is 6 (protons), and the mass number is 12 (6 protons + 6 neutrons).

State Pauli's exclusion principle and Hund's rule.

Answer: Pauli's exclusion principle states that no two electrons in an atom can have the same set of four quantum numbers. Hund's rule states that electrons occupy orbitals singly with parallel spins before pairing up.

What is the significance of quantum numbers?

Answer: Quantum numbers describe the energy, shape, orientation, and spin of an electron in an atom. They help determine the electron's position and behavior in the atomic structure.

Draw the shapes of s and p orbitals.

Answer: The s orbital is spherical, symmetrical around the nucleus. The p orbital is dumbbell-shaped, with two lobes along the x, y, or z axis.

Write the electronic configuration of Chromium (atomic number 24).

Answer: Chromium's electronic configuration is $[\text{Ar}] 3d^5 4s^1$. This is an exception to the Aufbau principle due to the stability of half-filled 3d orbitals.

Define ionic bond and covalent bond with one example each.

Answer: An ionic bond is formed by the transfer of electrons between atoms, e.g., NaCl (sodium donates an electron to chlorine). A covalent bond is formed by the sharing of electrons, e.g., H_2 (two hydrogen atoms share electrons).

Unit 2: Solutions, Acids and Bases (CO1)

Define molarity and normality. Write their formulas.

Answer: Molarity (M) is moles of solute per liter of solution, formula: $M = \frac{\text{moles}}{\text{volume (L)}}$. Normality (N) is equivalents of solute per liter of solution, formula: $N = \frac{\text{equivalents}}{\text{volume (L)}}$.

Explain the concept of pH.

Answer: pH measures the acidity or basicity of a solution, defined as the negative logarithm of hydrogen ion concentration: $\text{pH} = -\log[\text{H}^+]$. A pH below 7 is acidic, 7 is neutral, and above 7 is basic.

Calculate the pH of a solution with $[\text{H}^+] = 1 \times 10^{-5} \text{ M}$.

Answer: $\text{pH} = -\log[\text{H}^+] = -\log(1 \times 10^{-5}) = 5$. The solution has a pH of 5, indicating it is slightly acidic.

Define buffer solution. Give one example.

Answer: A buffer solution resists changes in pH when small amounts of acid or base are added. An example is a mixture of acetic acid (CH_3COOH) and sodium acetate (CH_3COONa).

State the limitations of Arrhenius theory of acids and bases.

Answer: Arrhenius theory only applies to aqueous solutions and defines acids as H^+ producers and bases as OH^- producers. It fails to explain acidity or basicity in non-aqueous solvents or substances like NH_3 .

Define solution. Classify solutions based on solubility.

Answer: A solution is a homogeneous mixture of a solute dissolved in a solvent. Based on solubility, solutions are unsaturated (can dissolve more solute), saturated (maximum solute dissolved), and supersaturated (excess solute dissolved).

Unit 3: Electrochemistry (CO2)

Define galvanic cell. Mention one application.

Answer: A galvanic cell is an electrochemical cell that converts chemical energy into electrical energy through spontaneous redox reactions. One application is in batteries, like a car battery.

State the applications of electrolysis.

Answer: Electrolysis is used in electroplating (coating metals), extraction of metals like aluminum, and production of chemicals like chlorine.

Distinguish between metallic conduction and electrolytic conduction.

Answer: Metallic conduction involves electron flow in metals, while electrolytic conduction involves ion movement in a solution or molten electrolyte. Metals conduct without decomposition, but electrolytes decompose during conduction.

List the types of batteries with one example each.

Answer: Types include primary batteries (e.g., dry cell), secondary batteries (e.g., lead-acid battery), and fuel cells (e.g., **hydrogen**-oxygen fuel cell).

State the applications of Li-ion batteries.

Answer: Li-ion batteries are used in mobile phones, laptops, and electric vehicles due to their high energy density and rechargeability.

Define electrolyte and non-electrolyte. Give one example each.

Answer: An electrolyte conducts electricity by dissociating into ions, e.g., $NaCl$. A non-electrolyte does not dissociate and does not conduct, e.g., sugar.

Unit 4: Corrosion (CO2)

List the factors that influence the rate of corrosion.

Answer: Factors include moisture, oxygen, temperature, presence of electrolytes (like salts), and metal surface condition.

Define rusting of iron.

Answer: Rusting of iron is the corrosion process where iron reacts with oxygen and water to form hydrated iron oxide ($Fe_2O_3 \cdot nH_2O$).

State two methods of preventing corrosion.

Answer: Two methods are applying protective coatings (e.g., paint) and using cathodic protection (e.g., sacrificial anode).

What is a sacrificial anode? Give one example.

Answer: A sacrificial anode is a more reactive metal used to protect another metal by corroding first. An example is zinc used to protect iron.

Explain the role of oxygen in corrosion.

Answer: Oxygen acts as an oxidizing agent in corrosion, accepting electrons from the metal to form oxides or hydroxides, like in rust formation.

Define corrosion and give one example.

Answer: Corrosion is the degradation of a metal due to chemical reactions with its environment. An example is the rusting of iron.

Unit 5: Water Treatment (CO3)

Define hard water. Mention two disadvantages.

Answer: Hard water contains high levels of calcium and magnesium ions, resisting soap lathering. Disadvantages include scale formation in boilers and reduced soap efficiency.

State two methods of softening hard water.

Answer: Two methods are the ion-exchange process and reverse osmosis.

Mention disadvantages of hard water used in industries.

Answer: It causes scale buildup in pipes and boilers, reducing efficiency, and increases soap consumption.

Define degree of hardness and its units.

Answer: Degree of hardness is the measure of calcium and magnesium ions in water. Its units are ppm (parts per million) and mg/L.

List the salts causing hardness of water.

Answer: Salts include calcium bicarbonate ($\text{Ca}(\text{HCO}_3)_2$), magnesium sulfate (MgSO_4), and calcium chloride (CaCl_2).

What is reverse osmosis in water treatment?

Answer: Reverse osmosis is a process where water is forced through a semipermeable membrane to remove dissolved salts and impurities, softening hard water.

Unit 6: Polymers & Engineering Materials (CO4)

1. Define polymerization. Give one example of addition polymerization.

- **Answer:** Polymerization is the process of joining monomers to form a polymer. An example of addition polymerization is the formation of polyethylene from ethylene.

2. State any three applications of nanotubes.

- **Answer:** Nanotubes are used in electronics (e.g., transistors), drug delivery, and as reinforcements in composite materials.

3. Define elastomer. Give one example.

- **Answer:** An elastomer is a polymer with elastic properties, capable of stretching and returning to its shape. An example is neoprene.

4. Write the uses of PVC.

- **Answer:** PVC is used in pipes, electrical insulation, and window frames due to its durability and resistance to chemicals.

5. What are liquid crystals? Give one example.

- **Answer:** Liquid crystals are materials with properties between liquids and solids, showing ordered molecular arrangement. An example is cholesterol benzoate.

6. Define composite materials. Give one example.

- **Answer:** Composite materials are made by combining two or more materials with different properties. An example is fiberglass.
-

Unit 7: Fuels (CO4)

1. **Write the composition and uses of CNG.**

Answer: CNG (Compressed Natural Gas) is mainly methane (CH_4). It is used as a fuel for vehicles and in power generation.

2. **Define fuel. Classify fuels based on occurrence.**

- **Answer:** A fuel is a substance that releases energy when burned. Fuels are classified as primary (e.g., coal) and secondary (e.g., petrol).

3. **Write the composition and uses of LPG.**

- **Answer:** LPG (Liquefied Petroleum Gas) is a mixture of propane (C_3H_8) and butane (C_4H_{10}). It is used for cooking and heating.

4. **State two advantages of hydrogen as a future fuel.**

- **Answer:** Hydrogen is abundant and produces only water when burned, making it eco-friendly.

5. **What is biogas? Mention one use.**

- **Answer:** Biogas is a mixture of methane and carbon dioxide produced from organic waste. It is used for cooking.

6. **State one disadvantage of hydrogen as a fuel.**

- **Answer:** Hydrogen is difficult to store and transport due to its low density and high flammability.
-

Unit 8: Environmental Studies (CO5)

1. **Define e-waste. Mention two sources.**

- **Answer:** E-waste is discarded electronic devices. Sources include old computers and mobile phones.

2. **Define deforestation. State one effect.**

- **Answer:** Deforestation is the clearing of forests for other uses. One effect is loss of biodiversity.

3. **What is the greenhouse effect?**

- **Answer:** The greenhouse effect is the trapping of heat in the Earth's atmosphere by gases like CO_2 , warming the planet.

4. **State two causes of water pollution.**

- **Answer:** Causes include industrial waste discharge and agricultural runoff containing fertilizers.

5. **Define green chemistry.**

- **Answer:** Green chemistry is the design of chemical processes to reduce environmental harm and waste.

6. **What is ozone layer depletion? Mention one cause.**

Answer: Ozone layer depletion is the thinning of the ozone layer in the atmosphere. One cause is the release of chlorofluorocarbons (CFCs).
